

Project: Implementing microservices with Code Pipeline and ECS

Course-end Project 2

Description

Objective

The objective of this project is to showcase the end-to-end process of deploying a microservices-based application with separate Backend (NodeJS) and Frontend (React) services using AWS CodePipeline and Amazon ECS. This project demonstrates the automation of CI/CD workflows, efficient containerization of both backend and frontend applications, and the deployment of these services on a scalable ECS cluster. The setup ensures high availability, seamless updates, and a reliable architecture utilizing serverless containers with AWS Fargate.

Problem Statement and Motivation

Real-time scenario:

DevBank, a rapidly expanding financial technology company, offers a digital banking platform to clients across North America. To enhance scalability and handle increasing user transactions, DevBank is transitioning its backend services (NodeJS API) and frontend interface (React application) to a microservices architecture deployed on AWS.

As a DevOps Engineer at DevBank, you are tasked with implementing automated CI/CD pipelines for both backend and frontend services. This project involves using Docker for containerization, AWS CodePipeline for automating builds, and Amazon ECS with AWS Fargate for deploying the services in a serverless, high-availability setup. Your responsibility is to ensure an efficient, automated deployment process that reduces release time and maintains a seamless user experience, enabling DevBank to scale its platform and deliver reliable services to its growing customer base.

Repositories

- [ReactFrontend](#)
- [NodeBackend](#)

Frontend buildspec.yml

```
version: 0.2

phases:
  pre_build:
    commands:
      - echo Logging in to DockerHub...
      - echo $DOCKERHUB_PASS | docker login -u $DOCKERHUB_USER --password-stdin
      - REPOSITORY_URI="docker.io/pradeepviswa/capstonefrontend"
      - CONTAINER_NAME="frontend-container"
      - IMAGE_TAG="latest"
  build:
    commands:
      - echo Build started on `date`
      - echo Building the Docker image...
      - docker build -t $REPOSITORY_URI:$IMAGE_TAG .
  post_build:
    commands:
      - echo Build completed on `date`
      - docker push $REPOSITORY_URI:$IMAGE_TAG
      - echo Writing image definitions file...
      - printf '["name":"$CONTAINER_NAME","imageUri":"%s"]' $REPOSITORY_URI:$IMAGE_TAG
> imagedefinition.json
artifacts:
  files: imagedefinition.json
```

Backend buildspec.yml

```
version: 0.2

phases:
  pre_build:
    commands:
      - echo Logging in to DockerHub...
      - echo $DOCKERHUB_PASS | docker login -u $DOCKERHUB_USER --password-stdin
      - REPOSITORY_URI="docker.io/pradeepviswa/capstonebackend"
      - CONTAINER_NAME="backend-container"
      - IMAGE_TAG="latest"
  build:
    commands:
      - echo Build started on `date`
      - echo Building the Docker image...
      - docker build -t $REPOSITORY_URI:$IMAGE_TAG .
  post_build:
    commands:
      - echo Build completed on `date`
      - docker push $REPOSITORY_URI:$IMAGE_TAG
      - echo Writing image definitions file...
      - printf '["name":"$CONTAINER_NAME","imageUri":"%s"]' $REPOSITORY_URI:$IMAGE_TAG
> imagedefinition.json
artifacts:
  files: imagedefinition.json
```

AWS Login

Region: us-east-1

AWS VPC



ECS Cluster

Name: vibrant-lion-otuntq (use the default generated name)

Infrastructure: Fargate Only (Serverless)

Amazon Elastic Container Service

Create cluster

Amazon Elastic Container Service

Favorites and recents New

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions

Account settings

AWS Resource Explorer 🔗

AWS Batch 🔗

Amazon ECR 🔗

Repositories 🔗

Online learning workshop 🔗

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Create cluster info

An Amazon ECS cluster groups together tasks, and services, and allows for shared capacity and common configurations. All of your tasks, services, and capacity must belong to a cluster.

Cluster configuration

Cluster name

productive-horse-mf9jmy

Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

▶ Service Connect defaults - optional

▼ Infrastructure - advanced info

Configure the manner of obtaining compute resources that will be used to host your application.

Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

☒ Fargate only

Serverless - you don't think about creating or managing servers. Great for most common workloads.

☐ Fargate and Managed Instances

Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.

☐ Fargate and Self-managed instances

Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.

▶ Monitoring - optional

Configure observability, encryption, and logging options to maintain compliance and operational visibility of your container environment.

▶ Encryption - optional

Choose the KMS keys used by tasks running in this cluster to encrypt your storage.

▶ Tags - optional info

Tags help you to identify and organize your clusters.

Cancel

Create

Create Task Definitions

[Amazon Elastic Container Service](#) > Task definitions

Create task definition for both frontend and backend:

1. container-port should be set to 3000
2. Container Detail:
 - For Frontend:
 1. Name: frontend-container
 2. URI: docker.io/pradeepviswa/capstonefrontend:latest

▼ Container - 1 [Info](#)

Container details

Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

frontend-container

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Essential container

Yes

Image URI

docker.io/pradeepviswa/capstonefrontend:latest

Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

[Browse ECR images](#)

Private registry [Info](#)

Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.

☐ Private registry authentication

Port mappings [Info](#)

Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
3000	TCP	container-port-protocol	HTTP	Remove

- For Backend:
 1. Name: backend-container
 2. URI: docker.io/pradeepviswa/capstonebackend:latest

▼ Container - 1 [Info](#)

Container details

Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

backend-container

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Essential container

Yes

Image URI

docker.io/pradeepviswa/capstonebackend:latest

Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

[Browse ECR images](#)

Private registry [Info](#)

Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.

☐ Private registry authentication

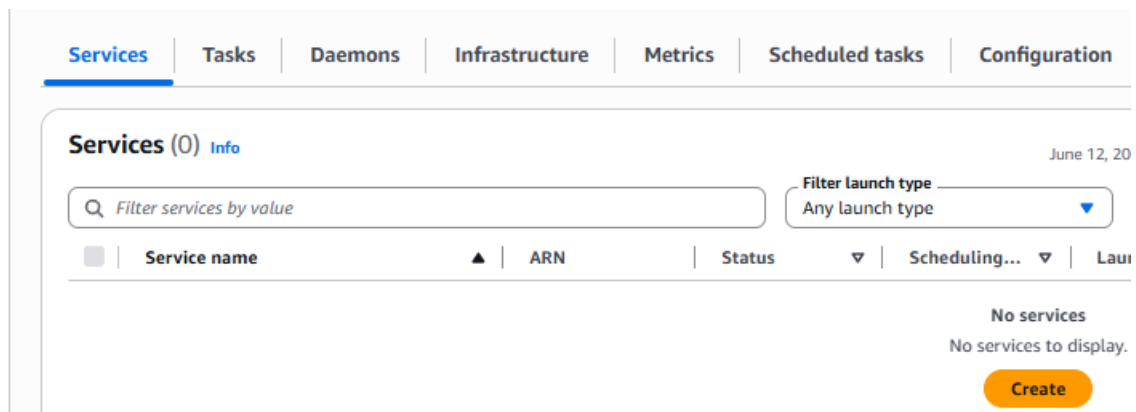
Port mappings [Info](#)

Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
3000	TCP	container-port-protocol	HTTP	Remove

Create ECS Service

Go to cluster (**vibrant-lion-otuntq**) > service



Create Service for both frontend and backend:

1. choose the task definition Family
2. Networking
 - Choose the previously created VPC
 - Choose all the public subnet
 - create new security group
 - inbound rule
 - Type = Custom TCP
 - Port range = 3000
 - Source = Anywhere

a. Note: you need to create the Security group only once, when you create the first service. Reuse the group in second service

Frontend service

Create service [Info](#)

Service details

Task definition family

Select an existing task definition family. To create a new task definition, go to [Task definitions](#).

frontend-container

Task definition revision

Latest

Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

Q 1

Service name

Assign a service name that is unique for this cluster.

frontend-container-service-wgglykpf

Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

Networking

VPC

[Info](#)

Select a VPC to use for your Amazon ECS resources.

vpc-0a5148af544001be9
vpc1

Create a new VPC

Subnets

Choose the subnets within the VPC that the task scheduler should consider for placement.

Choose subnets

Clear current selection

subnet-03494ce85fc8e4c89
subnet1
us-east-1c 10.0.0.0/28

Security group

[Info](#)

Choose an existing security group or create a new security group.

- ☒ Use an existing security group
☐ Create a new security group

Security group name

Choose an existing security group.

Choose security groups

sg-007e3b6a19820bb6d
default

Public IP

[Info](#)

Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

☒ Turned on

Backend Service

Create service

Info

Service details

Task definition family

Select an existing task definition family. To create a new task definition, go to [Task definitions](#).

backend-container

Task definition revision

Latest

Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

1

Service name

Assign a service name that is unique for this cluster.

backend-container-service-mwsvj7dd

Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

Networking

VPC

Info

Select a VPC to use for your Amazon ECS resources.

vpc-0a5148af544001be9

vpc1

Create a new VPC

Subnets

Choose the subnets within the VPC that the task scheduler should consider for placement.

Choose subnets

Clear current selection

subnet-03494ce85fc8e4c89

subnet1

us-east-1c 10.0.0.0/28

Security group

Info

Choose an existing security group or create a new security group.

Use an existing security group

Create a new security group

Security group name

Choose an existing security group.

Choose security groups

sg-007e3b6a19820bb6d

default

Public IP

Info

Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

Turned on

Service list

Services

Tasks

Daemons

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

Services (2)

Info

Last updated June 12, 2026, 22:14 (UTC+5:30)

Manage tags

Update

Delete service

Create

Filter services by value

Filter launch type Any launch type

Filter scheduling strategy Any scheduling strategy

Filter resource management type Any resource management type

< 1 >

☐	Service name	ARN	Status	Scheduling...	Launch type	Task defini...	Deployments and tasks	Last deployment
☐	backend-container-service-mwsvj7dd	arn:aws:ecs:us-e	Active	Replica	-	backend-cont...	1/1 Tasks running	Completed View
☐	frontend-container-service-wgglykpf	arn:aws:ecs:us-e	Active	Replica	-	frontend-cont...	1/1 Tasks running	Completed View

Setup CodePipeline

Create two pipelines : FrontendPipeline and BackendPipeline

Source stage

Create Code pipeline for Backend

Developer Tools > CodePipeline > Pipelines > Create new pipeline > build custom pipeline

The screenshot shows the 'Choose creation option' step (Step 1 of 7) in the AWS CodePipeline console. The breadcrumb navigation is 'Developer Tools > CodePipeline > Pipelines > Create new pipeline'. On the left, a sidebar lists the steps: Step 1 (Choose creation option), Step 2 (Choose pipeline settings), Step 3 (Add source stage), Step 4 - optional (Add build stage), and Step 5 - optional. The main area is titled 'Choose creation option' with an 'Info' icon. Below the title, it says 'Step 1 of 7'. There is a 'Category' section with three radio buttons: 'Deployment', 'Continuous Integration', and 'Automation'. The 'Build custom pipeline' option is selected and highlighted with a blue border. At the bottom right, there are 'Cancel' and 'Next' buttons.

Pipeline settings

Name: FrontendPipeline

The screenshot shows the 'Choose pipeline settings' step (Step 2 of 7) in the AWS CodePipeline console. The breadcrumb navigation is 'Developer Tools > CodePipeline > Pipelines > Create new pipeline'. The main area is titled 'Choose pipeline settings' with an 'Info' icon. Below the title, it says 'Step 2 of 7'. The 'Pipeline settings' section includes: 'Pipeline name' (text input with 'FrontendPipeline', a note 'Enter the pipeline name. You cannot edit the pipeline name after it is created.', and a character limit 'No more than 100 characters'); 'Execution mode' (radio buttons for 'Superseded', 'Queued' (selected), and 'Parallel', with an 'Info' icon); 'Service role' (radio buttons for 'New service role' (selected, with subtext 'Create a service role in your account') and 'Existing service role' (with subtext 'Choose an existing service role from your account')); and 'Role name' (text input with 'AWSCodePipelineServiceRole-us-east-1-FrontendPipeline', a note 'Type your service role name', and a checked checkbox 'Allow AWS CodePipeline to create a service role so it can be used with this new pipeline'). Below this is an 'Advanced settings' section with a right-pointing triangle icon and the text 'Configure artifact store location, encryption settings, and pipeline variables for your pipeline.' At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

Source

GitHub (via OAuth app) > connect to Github

Add source stage [Info](#)

Step 3 of 7

Source

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via OAuth app) ▼

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

[Connect to GitHub](#)

ⓘ The GitHub (via OAuth app) action is not recommended

The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. [Learn more](#)

☒ Enable automatic retry on stage failure

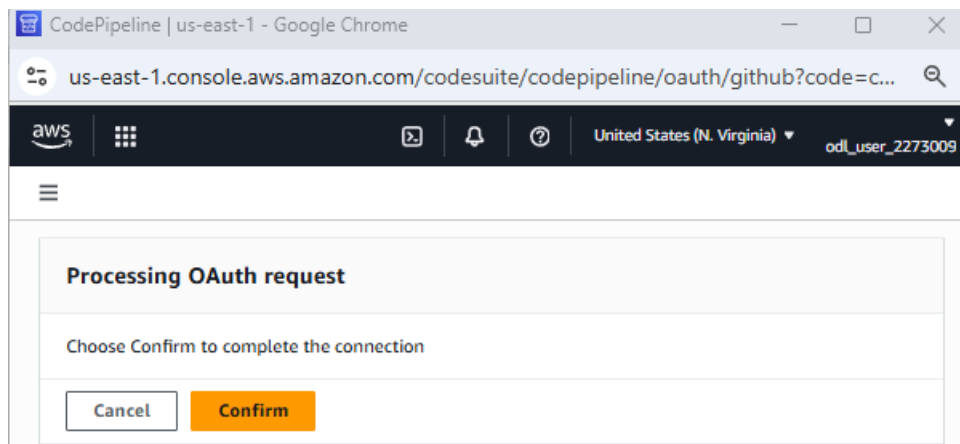
[Cancel](#)

[Previous](#)

[Next](#)

Connect with github

Authenticate



Select repository and branch

Add source stage [Info](#)

Step 3 of 7

Source

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via OAuth app) ▼

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

Connected

✓ You have successfully authenticated your account.



The GitHub (via OAuth app) action is not recommended

The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. [Learn more](#)

Repository

pradeepviswa/ReactFrontend



Branch

master



☒ Enable automatic retry on stage failure

Cancel

Previous

Next

Build

Build provider: Other build provider

AWS CodeBuild

Create Project

Add build stage - optional [Info](#)
Step 4 of 7

Build - optional

Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild ▼

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

or

Create project [↗](#)

 You must select a project

Project name: frontendProject

[Developer Tools](#) > [CodeBuild](#) > [Build projects](#) > Create build project

 **Continue to CodePipeline**
Create a new CodeBuild build project and return to CodePipeline to finish configuring your pipeline.

Create build project

Project configuration

Project name

frontendProject

A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.

add two environment variables under “Additional configuration”:

- i. “DOCKERHUB_USER” value = *your dockerhub user* type = plaintext
- ii. “DOCKERHUB_PASS” value = *your dockerhub password* type = plaintext

Environment variables		
Name	Value	Type
DOCKERHUB_USER	pradeepviswa	Plaintext
DOCKERHUB_PASS	██████████	Plaintext

[Add environment variable](#) [Remove](#) [Remove](#)

Buildspec: Use a buildspec file

▼ Buildspec

Build specifications

☐ Insert build commands
Store build commands as build project configuration

☒ Use a buildspec file
Store build commands in a YAML-formatted buildspec file

Buildspec name - *optional*

By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).

buildspec.yml

Build - *optional*

Build provider

Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild ▼

Project name

Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

🔍 frontendProject ✕

or

Create project ↗

☐ Define buildspec override - *optional*

Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

✔ Successfully created frontendProject in CodeBuild.

✕

Environment variables - *optional*

Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#) ↗

Add environment variable

Build type

☒ Single build

Triggers a single build.

☐ Batch build

Triggers multiple builds as a single execution.

Region

United States (N. Virginia) ▼

Input artifacts

Choose an input artifact for this action. [Learn more](#) ↗

SourceArtifact ✕

Defined by: Source

☒ Enable automatic retry on stage failure

Cancel

Previous

Skip build stage

Next

Build - *optional*

Build provider

Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild ▼

Project name

Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

🔍 frontendProject ✕

or

Create project ↗

☐ Define buildspec override - *optional*

Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

✔ Successfully created frontendProject in CodeBuild.

✕

Environment variables - *optional*

Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#) ↗

Add environment variable

Build type

☒ Single build

Triggers a single build.

☐ Batch build

Triggers multiple builds as a single execution.

Region

United States (N. Virginia) ▼

Input artifacts

Choose an input artifact for this action. [Learn more](#) ↗

SourceArtifact ✕

Defined by: Source

☒ Enable automatic retry on stage failure

Cancel

Previous

Skip build stage

Next

Test Stage : *Skip*

Add test stage - *optional* [Info](#)

Step 5 of 7

Test - *optional*

Test provider

Choose how you want to test your application or content. Choose the provider, and then provide the configuration details for that provider.

☒ Enable automatic retry on stage failure

[Cancel](#)

[Previous](#)

[Skip test stage](#)

[Next](#)

Deploy stage

Deploy provider: Amazon ECS

Input artifacts: BuildArtifact

Cluster name: ECS cluster name

Service name: frontend-service

Image definitions file : “builddefinition.json” (for frontend pipeline)

Add deploy stage [Info](#)

Step 6 of 7

Deploy - *optional*

Deploy provider

Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.

Amazon ECS

Region

United States (N. Virginia)

Input artifacts

Choose an input artifact for this action. [Learn more](#)

BuildArtifact

Defined by: Build

No more than 100 characters

Cluster name

Choose a cluster that you have already created in the Amazon ECS console. Or create a cluster in the Amazon ECS console and then return to this task.

Q vibrant-lion-otuntq

Service name

Choose a service that you have already created in the Amazon ECS console for your cluster. Or create a new service in the Amazon ECS console and then return to this task.

Q frontend-container-service-wgglykpf

Image definitions file - *optional*

Enter the JSON file that describes your service's container name and the image and tag.

builddefinition.json

Deployment timeout - *optional*

Enter the timeout in minutes for the deployment action.

|

☒ Configure automatic rollback on stage failure

☐ Enable automatic retry on stage failure

Cancel

Previous

Skip deploy stage

Next

Build pipeline

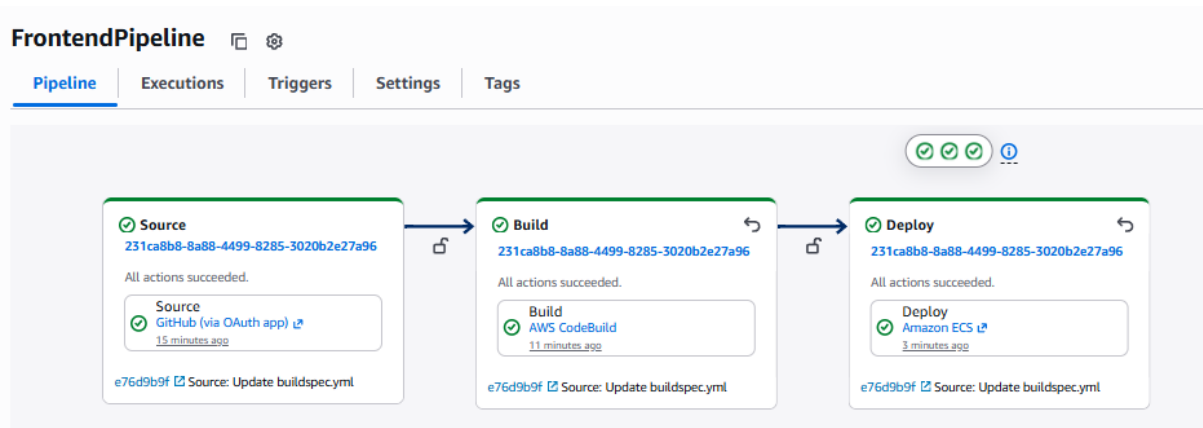


Image has been uploaded to docker-hub

[Repositories](#) / [capstonefrontend](#) / [General](#)

pradeepviswa/capstonefrontend

Last pushed less than a minute ago • Repository size: 951.2 MB • ☆0 • ↓67

Add a description  

Add a category  

[General](#)

[Tags](#)

[Image Management](#)

[Collaborators](#)

[Webhooks](#)

[Settings](#)

Tags

 DOCKER SCOUT INACTIVE
[Activate](#)

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
 latest		Image	less than 1 day	less than a minute

Web page

Go to cluster > service > service name (frontend-container-service-4fcey4yx)

Amazon Elastic Container Service

Favorites and recents [New](#)

[Express Mode](#)

[Clusters](#)

[Namespaces](#)

[Task definitions](#)

[Daemon task definitions](#)

[Account settings](#)

[AWS Resource Explorer](#)

[AWS Batch](#)

[Amazon ECR](#)

[Repositories](#)

[Online learning workshop](#)

[Documentation](#)

splendid-bat-zlwfja

Cluster overview

Cluster ARN

 [arn:aws:ecs:us-east-1:899823689630:cluster/splen:](#)

Services

Active: 2 | Draining: -

[Services](#)

[Tasks](#)

[Daemons](#)

[Infrastr](#)

Services (2) [Info](#)

 *Filter services by value*

- | | | |
|--------------------------|---|--|
| ☐ | Service name |  |
| ☐ | backend-container-service-0pkm1i5y | |
| ☐ | frontend-container-service-4fcey4yx | |

Task > click on task link

frontend-container-service-4fcey4yx [Info](#)

Service overview [Info](#)

Status
✓ Active

Tasks (1 Desired)

Health and metrics

Tasks

Logs

Deployments

Events

Config

Tasks (1/1)

Filter tasks by property or value

Filter
Any

Task	Last status	Desired status
8deffad9f2d04f68be175007	✓ Running	✓ Running

Copy public IP

Configuration

Operating system/Architecture
Linux/x86_64
CPU | Memory

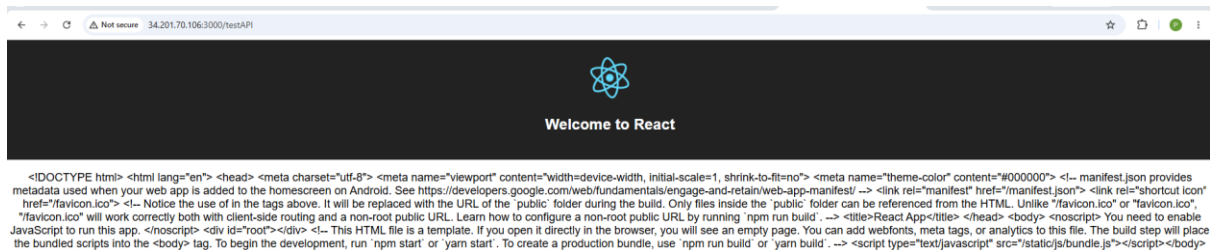
Capacity provider
Fargate
Launch type

ENI ID
eni-013ec37a4d6f7cbba
Network mode

Public IP
34.201.70.106 [open address](#)
Private IP

Web page URL

<http://34.201.70.106:3000/testAPI>



Create Code pipeline for Backend

Most of the steps are same as frontendPipeline, only difference screen shots mentioned below

Pipeline name: BackendPipeline

Choose pipeline settings [Info](#)

Step 2 of 7

Pipeline settings

Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.

No more than 100 characters

Add source stage [Info](#)

Step 3 of 7

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via OAuth app) ▼

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

Connected

✓ You have successfully authenticated your account. ✕

ⓘ The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. [Learn more](#)

Repository

pradeepviswa/NodeBackend ✕

Branch

master ✕

☒ Enable automatic retry on stage failure

[Cancel](#) [Previous](#) [Next](#)

Project Name: BackendProject

Developer Tools > CodeBuild > Build projects > Create build project

Continue to CodePipeline
Create a new CodeBuild build project and return to CodePipeline to finish configuring your pipeline.

Create build project

Project configuration

Project name

BackendProject

Create: "DOCKERHUB_USER" and "DOCKERHUB_PASS"

Environment variables

Name	Value	Type	
DOCKERHUB_USER	pradeepviswa	Plaintext	Remove
DOCKERHUB_PASS		Plaintext	Remove

[Add environment variable](#)

[Create parameter](#)

Buildspec

Build specifications

☐ Insert build commands
Store build commands as build project configuration

☒ Use a buildspec file
Store build commands in a YAML-formatted

Buildspec name - optional
By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, en example, buildspec-two.yml or configuration/buildspec.yml).

buildspec.yml

Build - *optional*

Build provider

Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild

Project name

Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

BackendProject



or

Create project

☐ Define buildspec override - *optional*

Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

✓ Successfully created BackendProject in CodeBuild.



Environment variables - *optional*

Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

Add environment variable

Build type

☒ Single build
Triggers a single build.

☐ Batch build
Triggers multiple builds as a single execution.

Region

United States (N. Virginia)

Input artifacts

Choose an input artifact for this action. [Learn more](#)

SourceArtifact
Defined by: Source

☒ Enable automatic retry on stage failure

Cancel

Previous

Skip build stage

Next

“imagedefinition.json”(for backend pipeline)

Add deploy stage [Info](#)

Step 6 of 7

Deploy - *optional*

Deploy provider

Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.

Amazon ECS

Region

United States (N. Virginia)

Input artifacts

Choose an input artifact for this action. [Learn more](#)

BuildArtifact
Defined by: Build

No more than 100 characters

Cluster name

Choose a cluster that you have already created in the Amazon ECS console. Or create a cluster in the Amazon ECS console and then return to this task.

vibrant-lion-otuntq

Service name

Choose a service that you have already created in the Amazon ECS console for your cluster. Or create a new service in the Amazon ECS console and then return to this task.

backend-container-service-mwsvj7dd

Image definitions file - *optional*

Enter the JSON file that describes your service's container name and the image and tag.

imagedefinition.json

Deployment timeout - *optional*

Enter the timeout in minutes for the deployment action.

☒ Configure automatic rollback on stage failure

☐ Enable automatic retry on stage failure

Cancel

Previous

Skip deploy stage

Next

Pipeline

aws

security group

Ask Amazon Q

Developer Tools

CodePipeline

Pipelines

BackendPipeline

Success

Conratulations! The pipeline BackendPipeline has been created.

BackendPipeline

Pipeline

Executions

Triggers

Settings

Tags

Source

0961be77-2690-4244-8cbe-7b96d3fd6c82

All actions succeeded.

Source

GitHub (via OAuth app)

7 minutes ago

a6c85d93 Source: Update buildspec.yml

Build

0961be77-2690-4244-8cbe-7b96d3fd6c82

All actions succeeded.

Build

AWS CodeBuild

6 minutes ago

a6c85d93 Source: Update buildspec.yml

Deploy

0961be77-2690-4244-8cbe-7b96d3fd6c82

All actions succeeded.

Deploy

Amazon ECS

1 minute ago

a6c85d93 Source: Update buildspec.yml

Docker image updated

hub

Explore

My Hub

Search

pradeepviswa

Your personal account

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / capstonebackend / General

pradeepviswa/capstonebackend

Last pushed 3 minutes ago • Repository size: 346.8 MB • 0 stars • 49 downloads

Add a description

Add a category

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

DOCKER SCOUT INACTIVE

Activate

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	3 minutes

Get backend public IP from ECS cluster

Amazon Elastic Container Service

Favorites and recents [New](#)

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions

Account settings

AWS Resource Explorer [↗](#)

AWS Batch [↗](#)

Amazon ECR [↗](#)

Repositories [↗](#)

Online learning workshop [↗](#)

splendid-bat-zlwfja

Cluster overview

Cluster ARN
[arn:aws:ecs:us-east-1:899823689630:cluster/sp](#)

Services
Active: 2 | Draining: -

Services | Tasks | Daemons | Infr

Services (2) [Info](#)

☐ Service name ▲

☐ backend-container-service-0pkm1i5y

Configuration
Operating system/Architecture
Linux/X86_64

Capacity provider
Fargate

ENI ID
[eni-099d7d0af6ec95ef1](#) [↗](#)

Public IP
[52.91.231.121](#) [open address](#) [↗](#)

Web Page

IP is 52.91.231.121

URL: http://52.91.231.121:3000/testAPI

← → ↻

⚠ Not secure

52.91.231.121:3000/testAPI

API is working properly

Point Frontend to correct Backend

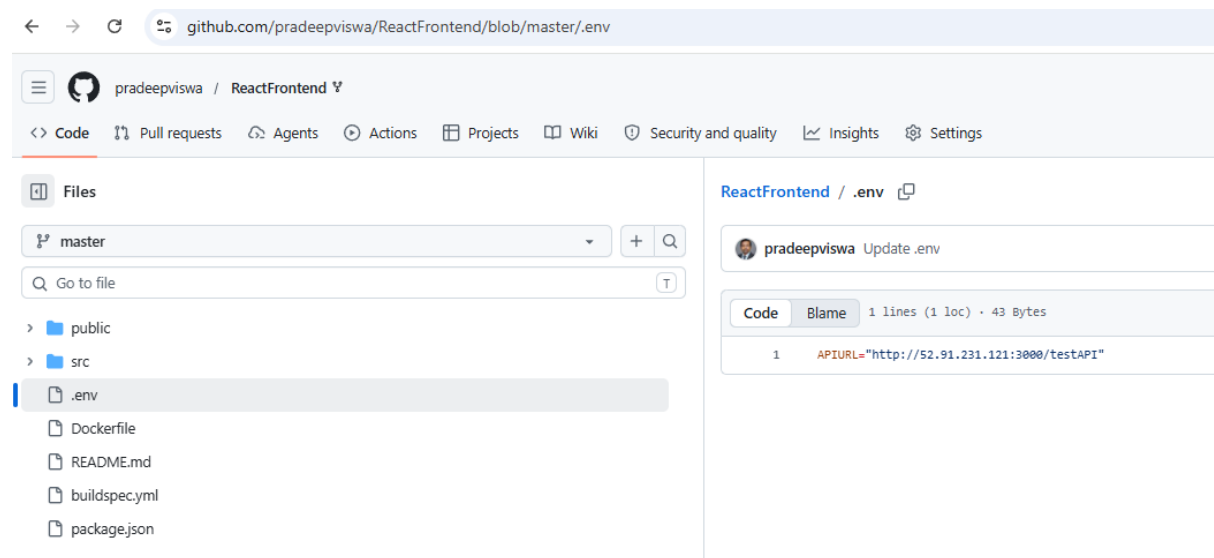
In frontend source code, we need to modify the .env file where the backend API URL will be updated. get the correct IP address of the backend task, and update the URL

1. File: ReactFrontend/.env

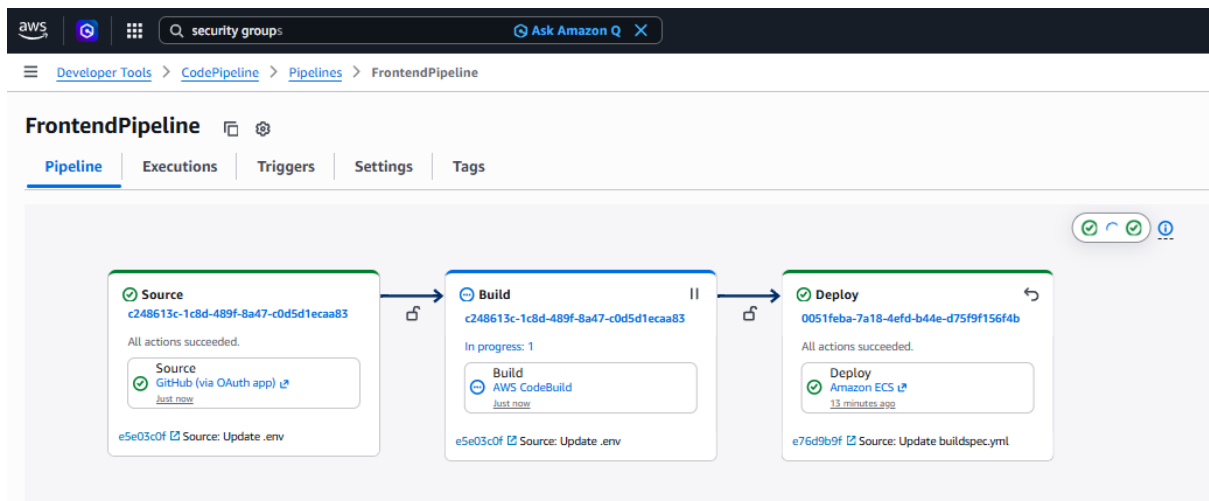
2. file content:

- a. `APIURL="http://<backend-IP-Addredd>:3000/testAPI"`

`http:// 52.91.231.121:3000/testAPI`



After commit build pipeline will trigger again, once done, browse frontend webpage again



After new build, public ip of frontend has changed

Configuration			
Operating system/Architecture Linux/X86_64	Capacity provider Fargate	ENI ID eni-0d92e31bd1d108318	Public IP copied 3.95.187.168 open address

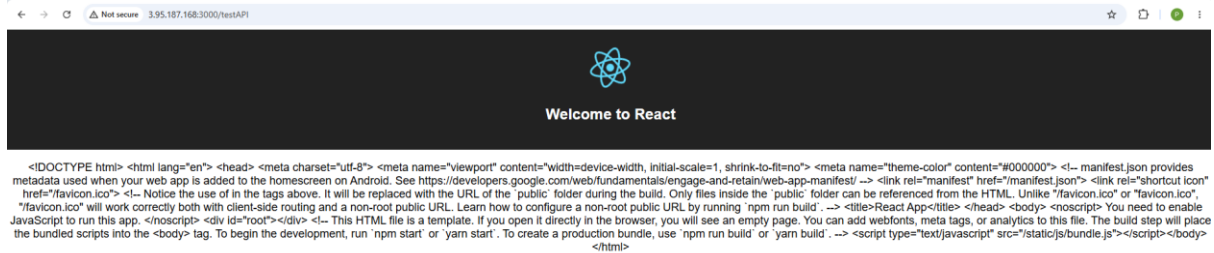
Browse frontend webpage again

URLs:

Frontend: <http://3.95.187.168:3000/testAPI>

Backend: <http://52.91.231.121:3000/testAPI>

Browse Frontend again



API communication is failing because of some bug in nodejs code which has to be handled by application developer team. From DevOps point of view, CI/CD pipeline is working as expected.